THE FRACTIONAL KNAPSACK PROBLEM

- We are given ‘n’ items and a knapsack.
- The item ‘i’ has a weight w_i and a profit p_i the knapsack has a capacity ‘M’.
- We are also allowed to take an item in fractional part.
- If a fraction x_i, $0 \le x_i \le 1$ of item i is placed into the knapsack then a profit of pi xi is earned.
- The objective is to fill the knapsack that maximizes the total profit earned.
- The ith item contributes the weight $w_i . x_i$ to the total weight in the knapsack and profit $p_i . x_i$ to the total profit.
- The problem is stated as:

$$\max \sum_{0 \le i < n} p_i x_i$$

$$s.t.$$

$$\sum_{0 \le i < n} w_i x_i \le M$$

$$0 \le x_i <= 1$$

$$p_i \ge 0, w_i \ge 0, 0 \le i < n$$

Algorithm

- If the items are already been sorted into non-increasing order of p[i] /w[i] then the algorithm given below obtains solutions corresponding to this strategy.

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Algorithm GreedyKnapsack (m, n)
// P[1 : n] and w[1 : n] contain the profits and weights
respectively of  items ordered so that p[i] / w[i] > p[i + 1] / w[i
+ 1].
// m is the knapsack size and x[1: n] is the solution vector.
{
  for i := 1 to n do x[i] := 0.0 // initialize x
  U := m;
   for i := 1 to n do
    {
     if (w[i] > U) then break;
    x [i] := 1.0; U := U –w[i];
    }
   if (i < n) then x[i] := U / w[i];
}
```

Example:

1. Consider the following instance of the knapsack problem: n = 3, m = 20, (P1, P2, P3) = (25, 24, 15) and (W1, W2, W3) = (18, 15, 10).
2. Find out the optimal solution for the following 3 items (P1, P2, P3) = (500,900,700) and (W1, W2, W3) = (1, 3, 7). Knapsack capacity is 5.